

Day 8 - IPv4 Addressing (pt 2)

Saturday, August 22, 2026

2:50 PM

8. IPv4 ADDRESSING : PART 2

MAXIMUM HOSTS PER NETWORK

Let's take a Class C Network:

192.168.1.0/24

(gives a range of 0 ---> 255)

Said another way, the HOST portion (the .0) is equal to 8 bits so...

Host portion = 8 bits = $2^8 = 256$

HOWEVER, since the Network Address (Network ID)

192.168.1.0 is Reserved

AND

192.168.1.255 (BROADCAST ADDRESS) is ALSO reserved.

The MAXIMUM Hosts per Network = $2^8 - 2 = 254$ hosts

What about a Class B Network ?

172.16.0.0/16 ----> 172.16.255.255/16

Host portion = 16 bits = $2^{16} = 65,536$

Maximum hosts per network = $2^{16} - 2 = 65,534$ hosts

What about a Class A Network ?

10.0.0.0/8 -----> 10.255.255.255/8

Host portion = 24 bits = $2^{24} = 16,777,216$

Maximum hosts per network = $2^{24} - 2 = 16,777,214$ hosts

THEREFORE:

The formula for calculating the number of HOSTS on a network is:

$2^N - 2$ (2 to the power of N - 2)

where N = number of HOST bits

FIRST / LAST USABLE ADDRESSES

Class C Network

192.168.1.0/24 (NETWORK ADDRESS)

Add 1 to the Host Portion = 00000001

192.168.1.1/24 = FIRST USABLE ADDRESS

192.168.1.255/24 (BROADCAST ADDRESS)

Subtract 1 from the BROADCAST ADDRESS = 11111110

192.168.1.254/24 = LAST USABLE ADDRESS

Class B Network

172.16.0.0/16 (NETWORK ADDRESS)

Add 1 to Host portion so 0000 0000 0000 0001

172.16.0.1/16 is the FIRST USABLE ADDRESS

172.16.255.255/16 (BROADCAST ADDRESS)

Subtract 1 to Broadcast Address so 1111 1111 1111 1110

172.16.255.254/16 is the LAST USABLE ADDRESS

Class A Network

10.0.0.0/8 (NETWORK ADDRESS)

Add 1 to Host portion so 00000000 00000000 00000001

10.0.0.1/8 is the FIRST USABLE ADDRESS

10.255.255.255/8 (BROADCAST ADDRESS)

Subtract 1 to Broadcast Address so 1111 1111 1111 1110

10.255.255.254/16 is the LAST USABLE ADDRESS

CISCO CLI DEVICE CONFIGURATION

R1> enable R1# show ip interface brief

Lists the Interfaces, IP Addresses, Method, Status, and Protocol.

Interfaces:

- What port interfaces are available/connected

IP Addresses

- Self explanatory. What IP Address is assigned.

Method

- What method was the IP address assigned?

Status (Layer 1 Status)

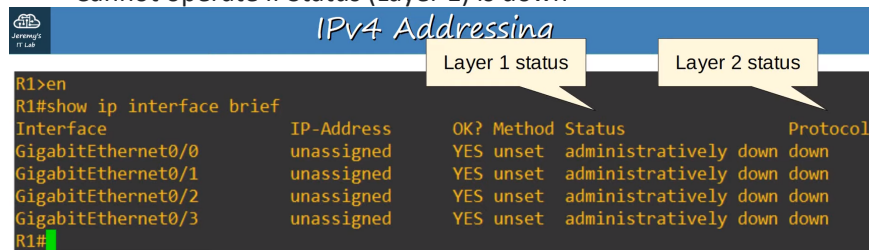
- Current status of interface
- 'administratively down' = Interface has been disabled with the 'shutdown' command

Administratively down is the DEFAULT status of Cisco Router interfaces.

Cisco Switch interfaces are NOT administratively down by DEFAULT.

Protocol (Layer 2 Status)

- Cannot operate if Status (Layer 1) is down



Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
GigabitEthernet0/3	unassigned	YES	unset	administratively down	down

- administratively down: Interface has been disabled with the 'shutdown' command.
- This is the default Status of Cisco router interfaces.
- Cisco switch interfaces are NOT administratively down by default.

// configure terminal cmd

R1# conf t

// This enters interface configuration mode

R1(config)# interface gigabitethernet 0/0

This can be shortened to 'g0/0' like they are listed in physical network maps.

```
R1(config-if)#ip address 10.255.255.254 ?
A.B.C.D IP subnet mask

R1(config-if)#ip address 10.255.255.254 255.0.0.0
R1(config-if)#no shutdown
R1(config-if)#
*Dec 7 08:29:08.937: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Dec 7 08:29:09.938: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R1(config-if)#
```

// This sets the IP ADDRESS and SUBNET MASK of device

R1(config-if) #ip address 10.255.255.254 255.0.0.0

// This enables the device

R1(config-if) #no shutdown

Two messages should appear showing the state has changed to 'up' (Status). Second message should show line protocol is now 'up' (Protocol).

// 'do' allows you to run a Privileged EXEC command from outside the mode.

R1(config-if) #do show ip interface brief

Good to confirm that the device/interface you have configured is up and running.

More 'show' CLI Commands

show interfaces [interface]

```
R1#show interfaces g0/0
GigabitEthernet0/0 is up, line protocol is up
  Hardware is IGBE, address is 0c1b.8444.f000 (bia 0c1b.8444.f000)
  Internet address is 10.255.255.254/8
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Auto Duplex, Auto Speed, link type is auto, media type is RJ45
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:06, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    167 packets input, 30159 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 0 multicast, 0 pause input
    350 packets output, 39097 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    105 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    1 lost carrier, 0 no carrier, 0 pause output
    0 output buffer failures, 0 output buffers swapped out
```

'show interfaces'

- Shows Layer 1 and Layer 2 information about the interface and some Layer 3.
- Shows MAC Address (or BIA address)
- IP Address
- ... and so much more

'show interfaces description'

- show description for interface.

Example:

// Configure mode for interface Gigabyte Interface 0/0

R1(config) #int g0/0

#####

R1(config) #description ## to SW1 ##

This sets the 'Description' column to display:

Interface Description

Gi0/0 ## to SW1 ##